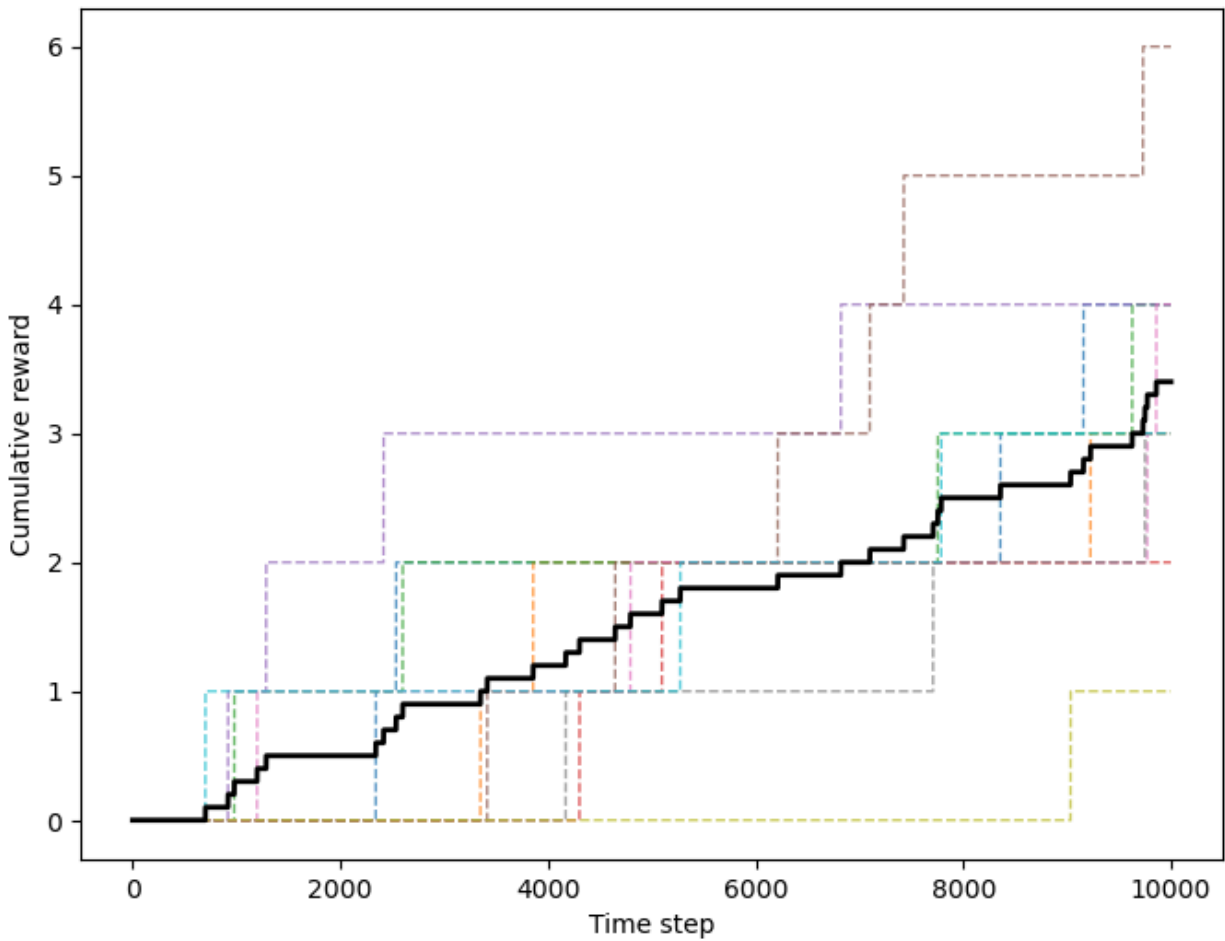


HW0 Written Answers

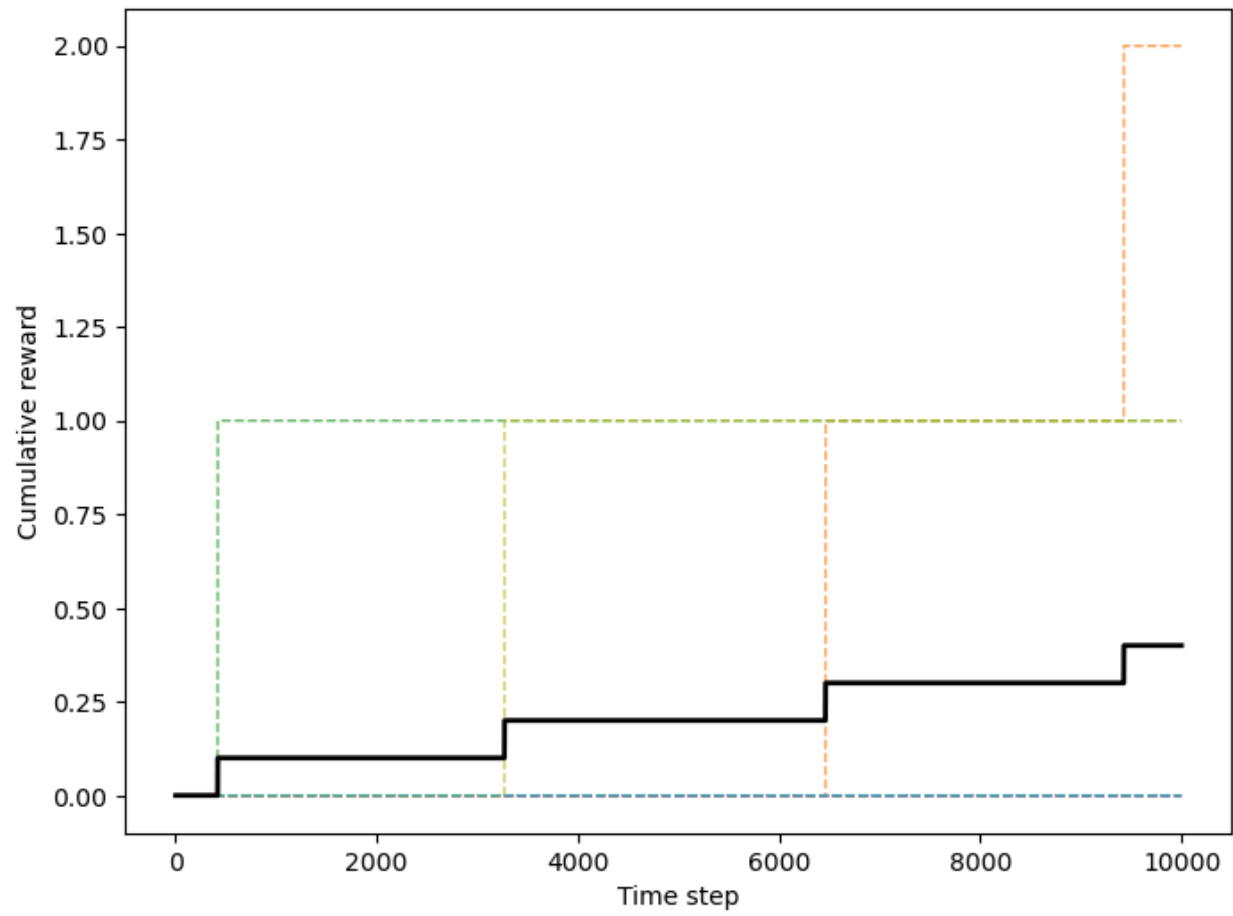
Adrian Ramirez

https://github.com/anramz29/reinforcement_learning/tree/main/HW1

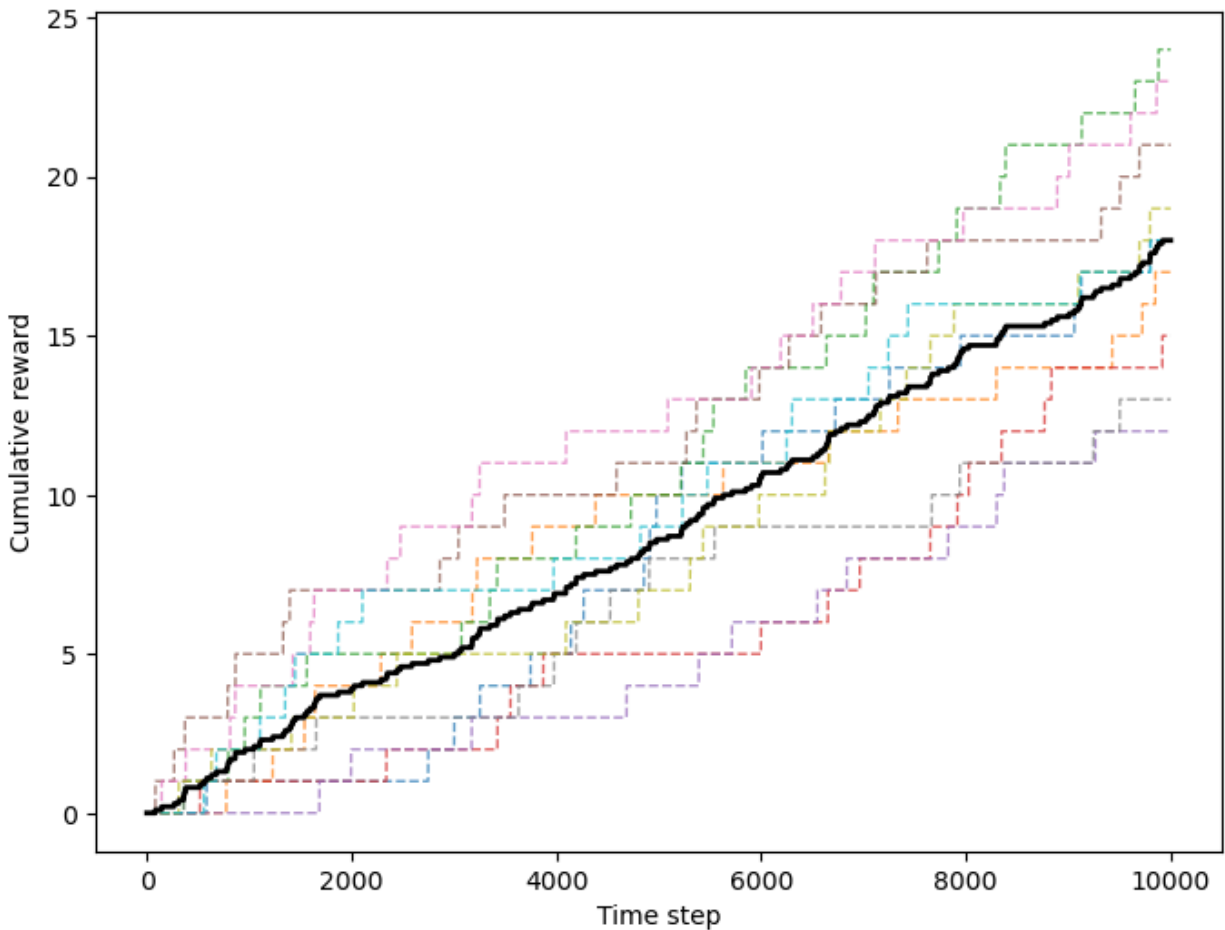
Random Policy:



Worse Policy:



Better Policy:



Written: Q3

The manual policy has significantly better performance than the random one because the person has full knowledge of the map (how to get through quadrants) with reasoning on how to get to the goal and correct errors from the perpendicular directions that happen 10% of the time. Whereas the automated policy acts without map knowledge, planning, or feedback to choose an optimal set in a given situation.

Written: Q4

1. The two policies
 - a. Worse Policy: I chose to give more weight to the “DOWN” and “LEFT” actions (30% probability respectively) in comparison to the “RIGHT” and “UP” actions (20% probability respectively). If the goal is (10, 10) with a start of (0, 0) then the

policy will generally spend more steps trying to push the agent to the bottom and left edges, leading to more wasted steps before it can get to the reward.

- b. Better Policy: I chose to give more weight to the “UP” and “RIGHT” actions (35% probability respectively) in comparison to the “DOWN” and “LEFT” actions (15% probability respectively). This encouraged the agent's behavior to be biased towards the goal, leading to a shorter number of steps and more instances of the agent getting to the reward and hence a larger cumulative reward.